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Review

Hallmarks of cancer—Then and now, and beyond

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SUMMARY

Cancer presents a remarkably instructive perturbation of mechanisms manifesting in our biology that have gone awry, eliciting a malady that is inexorably increasing in incidence and societal burden concomitant with healthier aging. The wealth of knowledge and data forthcoming from decades of cancer research can be organized into conceptually distinct but interconnected parametric dimensions that define the mechanistic foundation of the disease: aberrantly acquired functional capabilities (the hallmarks of cancer), enabling phenotypic characteristics, hallmark-conveying cells populating cancer microenvironments, and systemic interactions. Collectively, they provide a logical framework with which to illuminate the operating systems of these outlaw organs, from inception through multistage tumorigenesis to adaptive evolution. This review presents a concise synthesis of the hallmark conceptualization as it has been refined during the past 25 years, including a corollary hypothesis that mechanism-guided hallmark co-targeting could offer impactful new therapeutic strategies for treating human cancers.

INTRODUCTION

The "hallmarks of cancer" were conceived by Bob Weinberg and myself to help organize and reconcile the vast diversity of genetic and *phenotypic* characteristics observed in human cancers. We aimed to understand this complexity in the context of insights into how cancers arise via multistep pathways of tumorigenesis that lead to the formation of primary tumors, which subsequently evolve and progress due to selective advantage in the face of multifaceted barriers to continuing proliferative expansion, resulting, for example, in metastasis and adaptive resistance to therapy. We were guided by functional genetic studies in cultured cancer cells and model organisms and by increasingly sophisticated technologies for analyzing human tumors. The original perspective, published 26 years ago, posited that there were six hallmarks of cancer—acquired functional capabilities—consisting of sustaining proliferative signaling, inactivating growth suppressors, resisting programmed cell death, establishing replicative immortality, inducing or accessing vasculature, and activating invasion and metastasis.[1] An update in 2011 added two more hallmark capabilities, deregulating cellular metabolism and evading immune destruction,[2] and a ninth, unlocking phenotypic plasticity, was introduced in 2022,[3] which are now consolidated in Figure 1. A corollary in all three perspectives was an emphasis on the realization that transformed, mutant cancer cells did not in and of themselves define the biology of cancer, which rather depended upon the recruitment and reprogramming of heterotypic normal cells—accessories to the crime[4]—that contributed to the acquisition of hallmark capabilities.[1–4] An additional concept, integral to all three perspectives, pertained to abnormalities—enabling phenotypic characteristics—in incipient cancer cells and the tissue milieus wherein they were developing that serve to facilitate the acquisition of hallmark capabilities. Two such enabling phenotypic characteristics were initially introduced, namely loss of genomic integrity[1] and tumor-promoting inflammation.[2]

While broadly embraced by the cancer research community for its simplicity and utility in offering common principles underlying the daunting complexity of the disease, other frontiers of cancer research were suggesting additional parameters to the hallmarks conceptualization that might have general applicability to this heuristic tool. Potential new facets to the hallmarks' conceptualization were therefore introduced in the current decade in three perspectives.[3,5,6] This review intends to synthesize the concepts variously introduced in these six publications into a concise synopsis, and as such will not serve as a comprehensive review but rather a strategic overview, with citations of in-depth, focused reviews aimed to guide interested readers to the mechanistic details and supporting evidence and in turn the primary publications that laid the foundation for the various parameters described in essence herein.

As the hallmarks of cancer conceptualization evolved from 2000 to 2011 to encompass eight acquired functional capabilities and two enabling phenotypic characteristics that were then embellished with provocative new parameters in this decade, its attractive simplicity has arguably become more complicated. There is, however, the possibility to integrate the old and the new, then and now, into three qualitatively distinct sets of parameters, or conceptual dimensions (Figure 2). The first dimension integrates the hallmarks (Figure 1). The second

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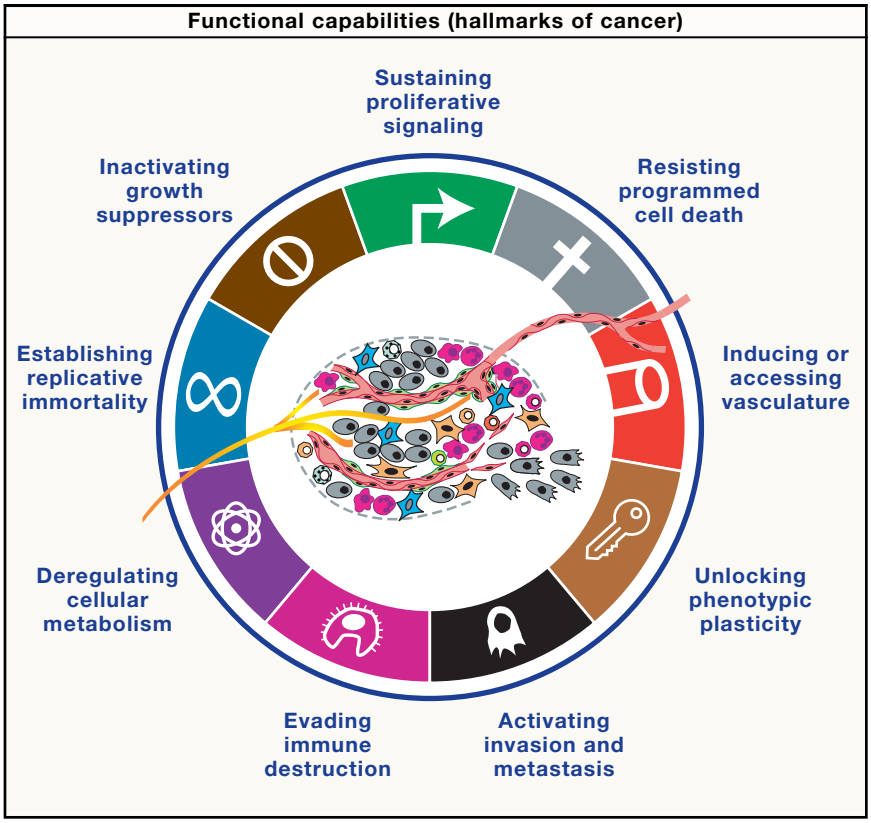


Figure 1. The hallmarks of cancer: Acquired functional capabilities

In the current synopsis, nine discrete capabilities are variously acquired at different stages of multistep tumor development and evolution in distinctive forms of cancer. Their acquisition is not strictly linear, and dependencies on some capabilities may be less than for others, or not at all, during particular pathways of tumorigenesis.

incorporates the now five hallmark-enabling phenotypic characteristics, while the third elaborates the growing roster of heterotypic cells that variously contribute to these functional capabilities. The fourth embodies multifaceted systemic interactions with the body in which a cancer arises (Figure 2), elaborated in a recent review.[6] The sections and subsections below describe parameters operative within each of these conceptual dimensions and beyond.

THE HALLMARKS OF CANCER: ACQUIRED FUNCTIONAL CAPABILITIES

The continuing premise is that the functional capabilities described below, acquired during multistage tumorigenesis and subsequent tumor evolution, are (1) distinctive in their mechanistic operations and contributions to the disease and (2) necessary—to varying degrees—in most forms of human cancer (Figure 1).

Sustaining proliferative signaling

As a fundamentally expansive disease, sustaining proliferative signaling of cancer cells is an obvious hallmark capability, one that is deeply embedded in our knowledge base about cancer. A majority of cancers contain activating mutations or chromosomal rearrangements in genes—dubbed oncogenes—that serve to chronically sustain otherwise transitory cell proliferation via triggering and stimulating progression through the cell division cycle. Most prominent is *KRAS*, which—via activating point mutations—upregulates a complex intracellular signaling network that, among other things, drives chronic proliferation. *KRAS* is mutated in ~30% of human tumors, including pancreatic adenocarcinoma (90%), colorectal cancer (50%), and lung adenocarcinoma (35%). Beyond *KRAS* and its homologs *NRAS* and *HRAS*, other commonly mutated proliferation-inducing "driver" oncogenes include *PIK3CA*, *BRAF*, and *BCR-ABL*. Other genes can become oncogenes by virtue of gene amplifications and other rearrangements, including often-circular extrachromosomal DNA elements (ecDNA), which collectively serve to induce/overexpress wild-type growth-stimulating proteins. Examples include *MYC*, *CCND1*, *EGFR*, and *EGFR2*/*HER2*, although the latter two can also be mutationally activated in certain cancer types. Most notably, the *MYC* gene, encoding a heterodimeric transcription factor (TF) that regulates thousands of genes, is amplified/rearranged, or its antagonists deleted, in 40% of human tumors. Other tumors lacking structural changes in the *MYC* gene locus have evidence for increased MYC expression and transcriptional activity by upregulation of its expression or post-translational modifications. Currently, 193 genes have been defined as proliferation-driving oncogenes by virtue of mutational alteration in human tumors and functional validation in model systems, as reported in the COSMIC database (Catalog of Somatic Mutations in Cancer) (Table S1). The precise mechanistic contributions of these non-canonical mutant genes to this and other hallmark capabilities are diverse and the subject of continuing investigation.

Pharmacological inhibition in cancer patients combined with pharmacological and genetic perturbations in mouse models of cancer has further validated and illuminated the roles and importance of proliferation-driving oncogenes in tumorigenesis and subsequent tumor progression.

Notably, both RAS and MYC, and their cooperative effects, have been implicated in stimulating multiple other hallmark capabilities above and beyond proliferative signaling. Interestingly, however, a small subset of human cancers, notably including some pediatric cancers, do not contain identifiable driving oncogenes that are recurrently mutated or amplified and are evidently governed by epigenetic reprogramming involving autocrine/paracrine supply of growth factors to sustain proliferative signaling. Most cancers, however, are anchored by genetic

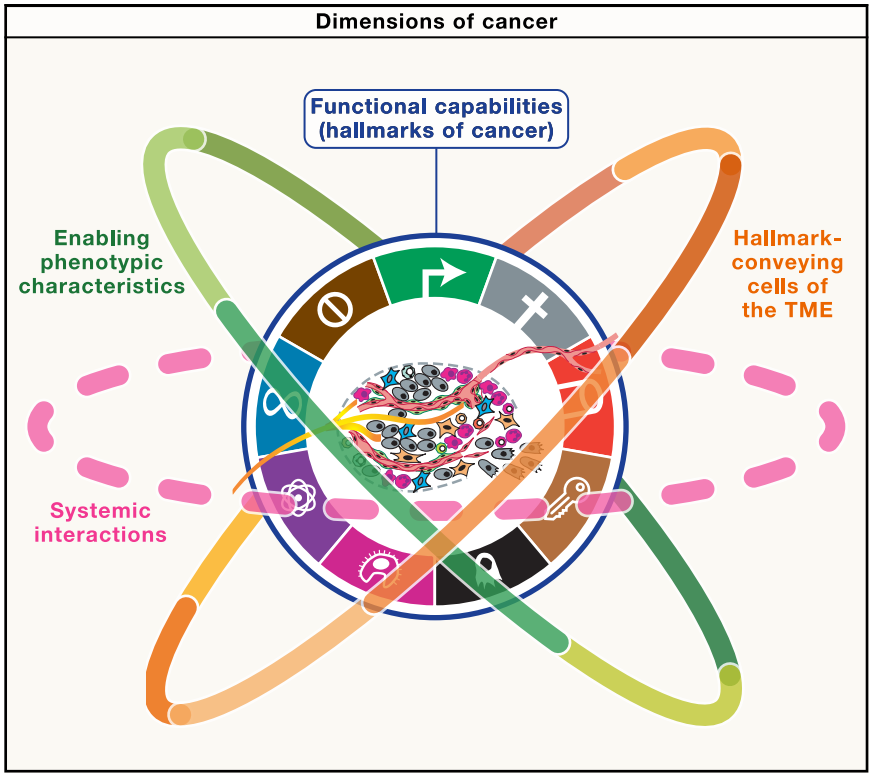


Figure 2. Dimensions of cancer
Four distinctive conceptual dimensions provide a logical basis for distilling the complexities of cancer into mechanistic components that collectively describe the essence of the disease. The parameters operative in each dimension are delineated throughout this review.

aberrations conveying the hallmark capability for sustaining chronic cell proliferation.[7–17]

Inactivating growth suppressors

The second hallmark capability is intuitively complementary to the first, given that the carefully orchestrated proliferation of normal cells during development and homeostasis is controlled not only by transitory proliferative signals but also by negative feedback loops and other mechanisms that activate genes encoding proteins that suppress cell proliferation. Both cell-intrinsic and -extrinsic mechanisms of growth suppression have been described, and these are variably inactivated or attenuated in fully developed cancers. Most prominent are inactivating mutations in a set of cancer-cell-intrinsic "tumor-suppressor genes" (TSGs) whose proteins normally act as gatekeepers for the sequential progression through the different phases—G1, G2, S, and M—of the cell division cycle. The roster includes a series of cyclin-dependent kinase inhibitors, *CDKN1A/B* (encoding p21 and p27) and *CDKN2A* (encoding p16INK4a and p14ARF) along with its paralogs *CDKN2B/C/D*, as well as the retinoblastoma protein (RB), which inhibits a transcriptional program necessary for cell-cycle progression, and the adenomatous polyposis coli protein (APC), which degrades a proliferation-inducing TF, β-catenin.

Triggering of an abnormality sensing and signaling hub, anchored by the TF TP53, can also indirectly shut off the cell cycle at the G1 and G2 transitions by upregulating *CDKN1A*, etc. in response to DNA damage and other cellular disruptions, including oncogenic signaling and hypoxia. Activation of TP53 also induces a plethora of other genes encoding modulators of cell proliferation, cell senescence, apoptotic cell death, non-proliferative differentiated states, and other hallmark parameters, as discussed below. Notably, mutations in TP53 are among the most frequent TSG mutations (~40%) across the spectrum of cancers analyzed by whole-genome sequencing. Interestingly, 60%–70% are missense mutations, centered upon a few hotspots that suggest particular importance for abrogating TP53's TSG functions. While loss-of-function and dominant negative mutations are well validated, there is evidence—albeit controversial—that some mutations convey oncogenic gains-of-function that have tumor-promoting effects above and beyond short-circuiting TP53's role as a tumor suppressor.

Currently, a collated roster of intrinsic growth suppressors lists 143 genes that have been implicated by their mutational inactivation in human cancers and filtered by functional validation in model systems (COSMIC) (Table S1). Less well-defined are paracrine tumor growth-suppressing mechanisms, including the long-recognized phenomenon of "contact inhibition," whereby neighboring epithelial cells and fibroblasts (and possibly other cell types) secrete or express on their cell surface signaling molecules that short-circuit the effects of growth signaling in adjacent, incipient cancer cells of origin. A second paracrine mechanism involves cell competition for limited supplies of nutrients and oxygen, etc., which can be inherently growth-inhibiting, such that cells stimulated to proliferate compete with each other to enable clonal outgrowth despite residing in naturally "tumor-suppressive" microenvironments. These mechanisms are variably circumvented during tumorigenesis, and other paracrine growth suppressors may be activated and then abrogated at later stages of tumor progression.

The mechanisms and importance of TSGs as barriers to tumorigenesis have been thoroughly established using experimental genetic (loss- and restoration-of-function) studies in engineered mouse models, by their prevalence in human tumors, and in some cases—notably involving CDKNs—by therapeutic targeting with drugs designed to replace the inhibitory functions of the abrogated TSGs. Collectively, it is clear that evading growth suppressors, principally by

mutational inactivation, is a fundamental acquired capability, one that is logically complementary to the stimulation of proliferative signaling by oncogenes.[10,18–29]

Resisting programmed cell death

A qualitatively distinct anti-proliferative cancer hallmark involves circumventing protective mechanisms that serve to eliminate aberrant or damaged cells—including cancer cells—via the induction of regulated pathways of cell suicide collectively termed "programmed cell death." Apoptosis is the prototype and most broadly operative form of programmed cell death across the spectrum of human cancers. In brief, indirect sensing of certain forms of cellular and genomic damage and abnormality can trigger either a paracrine (extrinsic) or an autocrine (intrinsic) program leading to apoptotic cell death. The extrinsic program involves signaling by secreted ligands from abnormality-sensing cells that activate "death receptors" of the tumor-necrosis family that in turn induce a cascade of degradative proteases (caspases) leading to cancer cell death. By contrast, the intrinsic program involves an increase in expression of so-called BH3-only proteins (PUMA, BIM, and NOXA) that act by binding to the effector proteins BAX and BAK at the mitochondrial outer membrane to initiate the apoptotic death pathway, which culminates in mitochondrial depolarization, caspase activation, and cellular disintegration. Levels of three gatekeepers, the anti-apoptotic proteins BCL-2, BCL-XL, and/or MCL-1, counterbalance the pro-apoptotic abnormality sensors, keeping cells alive by sequestering BH3-only proteins and their downstream effector proteins BAX and BAK. Upregulation of the gatekeepers is one mechanism by which cancer cells limit apoptotic cell death, as is mutational inactivation or gene deletion of *TP53*, whose loss otherwise triggers apoptosis by transcriptional upregulation of PUMA and NOXA.

While apoptosis is a prototypical barrier to aberrant hyperproliferation, it is now clear that other partially overlapping programmed cell death pathways can be activated in response to distinctive cellular abnormalities that arise in cancer cells. The roster of programmed cell death pathways includes necroptosis, ferroptosis, pyroptosis, and certain forms of autophagy and mitophagy, among which there is regulatory crosstalk and both shared and distinctive abnormality sensing and effector mechanisms with each other and with apoptosis. There is, in addition, the systemwide failure of cellular operations that results in cell death by ostensibly unregulated (non-programmed) cell lysis, termed necrosis. The importance of resisting these various mechanisms of cell death has been illuminated using functional genetics in model systems, further supported by associations of mutations in key components of these pathways in human cancers. In the case of apoptosis, further evidence comes from drug mimetics of the BH3-only proteins, which have been shown to trigger apoptosis in certain human and experimental tumors, with demonstrable therapeutic benefit. Notably, the BCL-2 inhibitor venetoclax is approved for treatment of chronic lymphocytic leukemia and acute myeloid leukemia, and it and other BH3-mimetics are being evaluated in broader indications. Currently, the clinical benefits of therapeutically disrupting resistance to apoptosis and other cell death programs across the spectrum of human cancers are unclear and under continuing investigation.

Interestingly, while defined as a tumor-suppressing barrier, apoptosis has been implicated in case studies as an "indirect" oncogene: apoptotic cancer cells can release signals that stimulate hallmark-enabling qualities in neighboring cancer cells and tumor-associated macrophages (TAMs), thereby facilitating tumorigenesis. Moreover, some apoptotic cells can escape from the program to die, emerging with partially degraded but still viable genomes that have acquired tumor-promoting oncogenic mutations. More generally, however, resisting programmed cell death is a necessary hallmark of cancer.[30–36]

Establishing replicative immortality

The fourth hallmark involves circumventing a sensing system that serves to prevent limitless, inappropriate cell proliferation, involving a cell division clock embedded in the concatenate of telomere repeats at the ends of the linear chromosomes. The "end replication problem" of DNA polymerases results in the loss of telomere repeats every time a cell duplicates its chromosomes and divides, thereby defining by their length the mitotic age of a cell. An elaborate assemblage of proteins referred to as the Shelterin complex binds to telomere repeats and serves as a sensor of mitotic age. When the declining number of telomere repeats reaches a set point indicative of excessive mitotic age, a cell-cycle arrest referred to as replicative senescence is triggered. Loss of TP53 and RB allows incipient cancer cells to overcome this checkpoint until further telomere loss leads to a second proliferative barrier termed crisis, which is characterized by severe chromosomal instability (CIN) and cell death, often involving autophagy. Cancer cells invariably disable this mitotic clock, in most cases by inducing expression of the catalytic subunit of the telomerase enzyme, the telomerase reverse transcriptase (TERT), which adds telomere repeats to chromosome ends. In humans and other long-lived animals (but not mice), TERT is repressed in differentiated cells, presenting a powerful tumor-suppressive mechanism.

In contrast to carcinomas, telomerase is only infrequently upregulated in cancer cells of mesenchymal and neuroepithelial origins. Instead, triggering of the anti-proliferative short telomere sensor is bypassed by inter-chromosomal recombination that extends the lengths of short telomeres, a mechanism referred to as "alternative lengthening of telomeres" (ALT). ALT activation involves changes in the telomeric chromatin structure as well as upregulation of telomeric repeat-containing long noncoding RNAs (TERRA).

Functional genetic studies in mouse models of cancer have established the necessity of avoiding this proliferative checkpoint for tumor growth and progression. Further testament comes from the observation that expression-enhancing mutations in the *TERT* gene promoter occur at an ~15% frequency on average across the spectrum of human tumors, and more so in glioblastoma and bladder cancer (~80%), melanoma (~60%), and selected others.

Notably, telomere erosion and dysfunction in the context of impaired DNA damage signaling during multistep tumorigenesis facilitate the acquisition of other hallmark capabilities by creating losses in genome integrity manifested as chromosomal fusions that break during mitosis to produce amplifications, deletions, and translocations before activation of telomerase (or ALT)

circumvents mitotic catastrophe to enable continuing cell viability and proliferative expansion. As such, the delayed activation of this capability is instrumental for tumorigenesis by inducing genome instability and mutational selection for expression and activity of hallmark-enabling genes. In sum, the realization that a mitotic clocking mechanism embedded in the telomeres is an important barrier to limitless cell proliferation during tumorigenesis has established its delayed abrogation—so as to establish replicative immortality—as a cancer hallmark.[37–44]

Beyond the capability for chronic proliferation of cancer cells

The first four hallmarks are intrinsic to the proficiency of cancer cells to proliferate chronically, as each involves corrupting and co-opting complementary mechanisms that otherwise serve to carefully orchestrate and limit cell proliferation during embryonic development, organogenesis, bodily growth, and systemic homeostasis. Tumors, however, are not bags of proliferating cancer cells. Rather, tumors are "outlaw organs" comprising multiple cell types that have coalesced in the context of phenotypic selection for expansive growth of neoplastic lesions during tumorigenesis.

As such, symptomatic tumors develop despite multi-faceted protective mechanisms involving not only these cancer cell-intrinsic barriers but also ones that are operative in the milieu within which clusters of incipient and progressing cancer cells reside, namely the so-called tumor microenvironment (TME), as well as the systemic macroenvironment of the organism. The acquired hallmark capabilities described below reflect other physiological barriers that are circumvented to produce cancers culminating in morbidity and mortality.

Inducing or accessing vasculature

The fifth hallmark embodies the intuitive necessity to supply cancer cells with oxygen and nutrients and evacuate waste products via access to blood vessels. This requirement is evident in the observation that nests of growing cells will become hypoxic and nutrient-deprived when the nearest capillary is more than 100 μm away, limiting continuing proliferative expansion and culminating in necrotic cell death. Additionally, the tumor vasculature has broader and more complex tumor-supporting roles, involving secretion of paracrine signaling molecules that modulate functions and phenotypic states of cancer and accessory cells in the TME. Much as in development, organogenesis, wound healing, and certain physiological situations, incipient tumors can demonstrably turn on angiogenesis, the growth of new blood vessels, via multiple mechanisms, mostly involving hypoxia-induced secretion of angiogenetic growth factors—such as VEGFA and Angiopoeitin 2 (ANGPT2), along with a plethora of other vascular regulatory factors—that signal through receptors expressed on vascular cells. Induction of angiogenesis with consequent neovascularization involves a complex sequence whereby quiescent endothelial cells become activated to proliferate and sprout from an existing vessel, followed by formation of tubes that fuse to form capillary-like loops capable of supporting blood flow. Concomitantly, peri-endothelial support cells—primarily pericytes—are induced to proliferate and envelop the neovasculature so as to stabilize it and support its functions. Notably, the neovasculature induced by chronic pro-angiogenic signaling in tumors is typically more permeable (leakier), chaotic, and comparatively dysfunctional, with sparser and more dissociated pericyte coverage. The angiogenic vasculature also presents an impediment to extravasating T and B lymphocytes, contributing to the immuno-evasive hallmark discussed below.

Tumor angiogenesis is switched on at varying points during multi-step tumorigenesis, and its importance has been revealed using functional genetics and pharmacological inhibition of pro-angiogenic signaling pathways. There is, in addition, another means by which incipient hyperplasias, primary tumors, and metastatic lesions can be supplied with oxygen and nutrients, and that is co-option of normal quiescent tissue by invasion along blood vessels in adjacent or distant normal tissues, without inducing angiogenesis. The importance of vascular co-option has been most clearly revealed by genetic and pharmacological inhibition of angiogenesis, which elicits invasive vascular co-option that enables continuing proliferative expansion. In sum, angiogenesis and vascular co-option are collectively established as fundamental components of the hallmark capability of tumors to become vascularized.[45–54]

Deregulating cellular metabolism (for proliferative energetics)

Another important requirement for expansive cell proliferation is ensuring a sufficient supply of energy and the cellular building blocks needed for cell growth, division, and survival. Non-proliferative or transitorily proliferative normal cells in their particular tissue ecosystems typically depend either on oxidative phosphorylation in mitochondria or, in special cases, on glycolysis, so as to produce energy from glucose metabolism. In marked contrast, cancer cells adopt complex metabolic phenotypes that are dynamic in space (within the TME) and in time (during tumorigenesis and subsequent progression). Firstly, many cancer cells are in essence metabolic hybrids, simultaneously employing both oxidative phosphorylation and aerobic glycolysis to varying degrees in order to optimize production of energy sources and cellular building blocks. Second, other sources of fuel beyond glucose are utilized, including paracrine secretion of lactate produced by aerobic glycolysis, which can be imported and utilized as an alternative source of fuel in a form of metabolic symbiosis, as can certain amino acids, most prominently glutamine, which are locally secreted (by TME cells) or blood-born.

The programming of such hybrid metabolic engines in cancer cells to support continuous cell proliferation is dictated both by cancer cell intrinsic factors, most notably mutations in oncogenes and tumor suppressors, as well as cell-of-origin, and by extrinsic factors in the TME, involving crosstalk with other cell types, including cancer-associated fibroblasts (CAFs), vasculature, macrophages, and T cells. Moreover, non-cellular attributes of the TME are instrumental, including gradients of hypoxia, acidosis, and varying local abundance of amino acids (glutamine, etc.), as well as molecules that are either secreted or released from extracellular sequestration, including cytokines and other proteins, and lipids. Moreover, all of these extrinsic factors and the consequently engaged metabolic pathways vary dynamically as tumors develop, progress, metastasize, and adapt to therapy. Autophagy and related cellular recycling programs can also be instrumental

in supporting tumor growth in conditions of nutrient insufficiency, in another manifestation of metabolic plasticity. Genetic and pharmacological perturbation of these various parameters has established their functional importance, as well as the inherent plasticity and adaptability of metabolic programming, posing important challenges for effective therapeutic targeting of this instrumental cancer hallmark.[6,55–62]

Unlocking phenotypic plasticity (and consequent functional heterogeneity)

It is now evident that the clonal hypothesis for cancer development and evolution, embedded in our conceptualization of cancer, is only partially correct. The advent of powerful technologies for profiling cells in tumors at the single-cell level has collectively revealed the widespread existence of intratumoral heterogeneity, driven not just by mutational selection but also by an acquired capability for phenotypic plasticity, which contributes in myriad ways to circumventing roadblocks and constraints faced during tumorigeneses, tumor growth, metastasis, and therapy resistance.

Two classes of plasticity are well recognized: one is *intralineage plasticity*, whereby cancer cells can (1) de-differentiate back along the developmental trajectories of their differentiated cells of origin in order to acquire tumor-promoting functions resident in such progenitor cells; (2) block differentiation of transformed progenitor cells that would otherwise be prone to enter irrevocable terminal differentiation pathways; or (3) acquire and/or shuttle back and forth between a spectrum of "cancer stem cell" (CSC) states that harbor varying degrees of tissue stem cell characteristics, dubbed "cancer stemness." Moreover, latent plasticity to enable tissue regeneration is endowed in many normal cell types and evidently awakened in cancer. Notably, recent studies indicate that TP53 activity helps maintain differentiated states, whereas its loss/attenuation enables de-differentiation and lineage plasticity.

A second class is "*trans*-lineage plasticity," involving *trans*-differentiation, wherein cancer cells are reprogrammed to adopt distinct cell types from different developmental lineages. The prototype is epithelial-mesenchymal plasticity (EMP), historically known as the epithelial-mesenchymal transition (EMT) that transpires developmentally during the programmed differentiation trajectories of embryogenesis and organogenesis. Unlike the discrete transition when a normal cell with an epithelial phenotype switches to a mesenchymal phenotype during development, cancer cells typically exhibit hybrid EMT phenotypes, involving dynamically varying degrees of epithelial and mesenchymal characteristics in the same cell and in populations of cells inside tumors. More generally, cell-state switching is now well documented beyond EMP, including transitions from classical (columnar) epithelial cells to squamous epithelial cells, to neuroendocrine cells, to vascular endothelial cells, and to senescent cells. While less precisely delineated, it seems likely that these broader cell state interconversions also involve a spectrum of hybrid phenotypes, in addition to complete cell-type switching. Moreover, some such conversions have been shown to be reversible, in the forms of a mesenchymal to epithelial transition (MET), an endothelial to epithelial transition, and transitions back out of senescent cell and CSC states.

Phenotypic plasticity has also been documented in most of the accessory cells of the TME, including dynamic sub-states of TAMs, tumor-associated neutrophils (TANs), and CAFs.

Both classes of phenotypic plasticity are diversely enabled by both mutational genetic and non-mutational epigenetic mechanisms—cancer-cell-intrinsic and microenvironmental—based in part on orchestrating the expression and activity of lineage-defining TFs, modulated by a variety of post-transcriptional regulatory modifications. The foundations are set by oncogenic mutations and cell type of origin, overlaid by changes in DNA methylation and modification of the "histone code" regulating accessibility of chromosomal domains, all converging on plasticity-inducing TF gene expression, whose regulation is further modulated by the local milieu enveloping the heterotypic cells populating tumors, involving gradients of hypoxia, metabolites, and secreted signaling molecules, as well as nutrient availabilities and insufficiencies. Notably, beyond genome instability, other enabling phenotypic characteristics can contribute to plasticity, e.g., tumor-promoting inflammation and innervation. Manifestations of the acquired capability for both forms of phenotypic plasticity include modulating other hallmark capabilities that are concordantly facilitating tumorigenesis and subsequent tumor evolution, as well as the frequent development of adaptive resistance to virtually all forms of cancer therapy, with the engaged mechanisms varying by tumor type and the particular forms of plasticity that are operative.[24,63–79]

Evading immune destruction

It is now integral to our conceptualization that cancers almost invariably avoid attack and immune destruction by acquiring the hallmark capability to suppress the adaptive and certain components of the innate immune systems that recognize and would otherwise react against neoplastic abnormalities. One aspect of immune evasion exploits the inborn programming of T cells and natural killer (NK) cells to be transiently and not chronically activated so as to minimize their potential to cause autoimmunity while fighting infections and other abnormalities. Activation—e.g., upon recognizing neoantigens—in draining lymph nodes and tumors triggers a temporal differentiation pathway in T cells, leading to an exhausted and dysfunctional state, thus limiting the duration of immune destruction (of tumors). Progression through the exhaustion pathway is stimulated by chronic tumor antigen exposure and orchestrated by the up- and downregulation of transcriptional regulatory circuits, which are in part hardwired and in part modulated by extrinsic factors.

While the limitations posed by the exhaustion pathway are implicit to T cell biology, the overarching importance of evading immune destruction is substantiated by the multi-faceted immunosuppressive mechanisms operative in the TME, as well as its capability to systemically impair T cell development in the lymphoid organs. Virtually every cell type in the TME has the demonstrable capability to suppress the recruitment and/or activity of CD8 and/or CD4 T cells, albeit to varying extents in different tumor types and at different stages of multistep tumorigenesis and subsequent tumor progression. While it is not surprising that developing cancer cells—in the context of their neo-Darwinian adaptation—elaborate immunosuppressive functions, so too can endothelial cells and pericytes of the tumor

vasculature, CAFs, innate immune inflammatory cells (substates of macrophages, neutrophils, and NK cells), and innervation, along with regulatory T and B cells (Treg and Breg). Moreover, the physical microenvironment can suppress T cell and NK cell-mediated immune responses in the forms of hypoxia, extracellular matrix composition, and interstitial pressure, along with the chemical milieu in the extracellular space, involving metabolic insufficiencies for T cell activity (amino acids, lipids, and sugars), in addition to the immunoregulatory factors (cytokines and chemokines, exosomes, etc.) produced by the various cell types populating the TME.

Collectively, illuminating and functionally validating these multi-faceted mechanisms of immunosuppression in experimental systems and clinical settings has crystalized the importance of evading immune destruction as a hallmark and spawned a new field of cancer therapeutics aiming to disrupt it, with proof of concept coming from the therapeutic benefits of immune checkpoint inhibitors (ICIs) that modulate the development and self-limiting exhaustion of neoplasia-attacking T cells.[46,48,80–85]

Activating invasion and metastasis

The acquired capability for invasion and metastasis is the most mechanistically complex hallmark, involving multiple steps of adaptation and survival beyond those operative in primary tumors, one that accounts for much of cancer morbidity and mortality. As noted above, one driving force for invasion and metastasis is gaining access to additional sources of vasculature to nourish growth beyond the physical and metabolic constraints of solid tumors. Much like multistep pathways leading to primary tumors, metastasis unfolds in a multistep pathway—dubbed the "metastatic cascade"—whereby cancer cells invade into blood or lymphatic vessels within or adjacent to primary tumors, flowing as circulating tumor cells that eventually extravasate as disseminated cancer cells into distant tissues.

The seeded cancer cells can either grow to colonize new microscopic and then macroscopic tumor nodules or lie dormant, with the possibility to eventually be awakened to proceed along the pathways of colonization. Dormancy of seeded cancer cells, which can last decades, has been variously associated with immune surveillance, failure to induce angiogenesis in such cases where invasive vascular co-option is impaired/inadequate, paracrine tumor suppression, and metabolic insufficiencies.

Many tumor types show preferential metastasis to particular organs wherein the microenvironment—intrinsically or by induction—is supportive of expansive growth. Indeed, an early step in the metastatic pathways for many tumors is the programming, via factors secreted into the circulatory system, of a supportive microenvironment, dubbed the "premetastatic niche," in distant tissues. Both genetic and non-genetic mechanisms underlay pathways of invasion and metastasis.

Perplexingly, although one could envision mutant metastasis genes that orchestrate the metastatic pathway much like mutated oncogenes drive primary tumorigenesis, there is abundant evidence arguing against a commonality of genes that are mutationally activated in metastatic cells to specifically regulate the steps in metastasis, although a few gene mutations—e.g., in *TP53*—are consistently associated with metastatic lesions. Rather, mutant oncogenes found in metastases are typically detected in their primary tumors, either as truncal mutations or within the intratumoral heterogeneity that is common to many tumors. There is often, however, a genetic component, involving new incidents of CIN leading to regional deletions and amplifications (somatic copy-number alterations) and whole genome doubling, as well as the appearance of new point mutations across the genome. Concordantly, higher incidences of *TP53* mutations are associated with many of the increases in genomic instability seen in metastases compared with primary tumors. The phenotypic consequences of CIN remain incompletely understood but are likely to facilitate many of the adaptive phenotypes required for multi-step pathways of metastasis.

Additionally, a variety of instrumental non-genetic mechanisms underlie the metastatic pathway, including phenotypic plasticity that enables EMP and other epigenetic regulatory changes, as well as metabolic plasticity to enable tumor growth in ectopic microenvironments. Moreover, paracrine stimulation by the chemical milieu released from TME cells in primary tumors and tissue-resident cells in locally invaded and distant metastatic sites can be influential in metastatic adaptation. Thus, metabolites, including lactate, succinate, glutamate, and certain amino acids, can help fuel growth of cancer cells that have adopted states of metabolic plasticity. Extracellular metabolites can also trigger pro-growth and pro-invasive signaling pathways via cell surface receptors or uptake transporters. Other signaling molecules—e.g., chemokines and cytokines—secreted from resident cells in invasive locales or metastatic niches adjacent to cancer cells can similarly instruct invasive growth and metastatic colonization, with the cancer cells reciprocally instructing supportive microenvironments. Biophysical parameters—pH, oxygen, matrix stiffness, and interstitial fluid pressure and flow—can also modulate components of this multifactorial capability. In addition to overcoming barriers to cell invasion and immigration embodied in normal tissues, thereby enabling and stimulating lesional outgrowth, these factors can variously contribute to suppressing or evading immune surveillance and attack. The daunting diversity of invasive and metastatic pathways across the spectrum of human cancers, and the capability of immigrating cancer cells to assume dormant states, present major strategic challenges—and opportunities—both to fully elucidate mechanisms and to develop effective mechanism-guided therapies that disrupt this instigator of most forms of clinically debilitating cancer.[86–94]

As was already appreciated in the original conceptualization—see Figure 4 of Hanahan and Weinberg[1]—hallmark capabilities are not necessarily acquired simultaneously but typically at different stages of tumorigenesis and tumor evolution, with the temporal order of their acquisition varying by tumor type and cell/tissue of origin and by the particular oncogenes driving incipient neoplasia and phenotypic progression.

HALLMARK-ENABLING PHENOTYPIC CHARACTERISTICS

While the hallmarks are defined as capabilities that are cooperatively functioning to orchestrate the disease, the means by which they are acquired is another dimension, reflecting

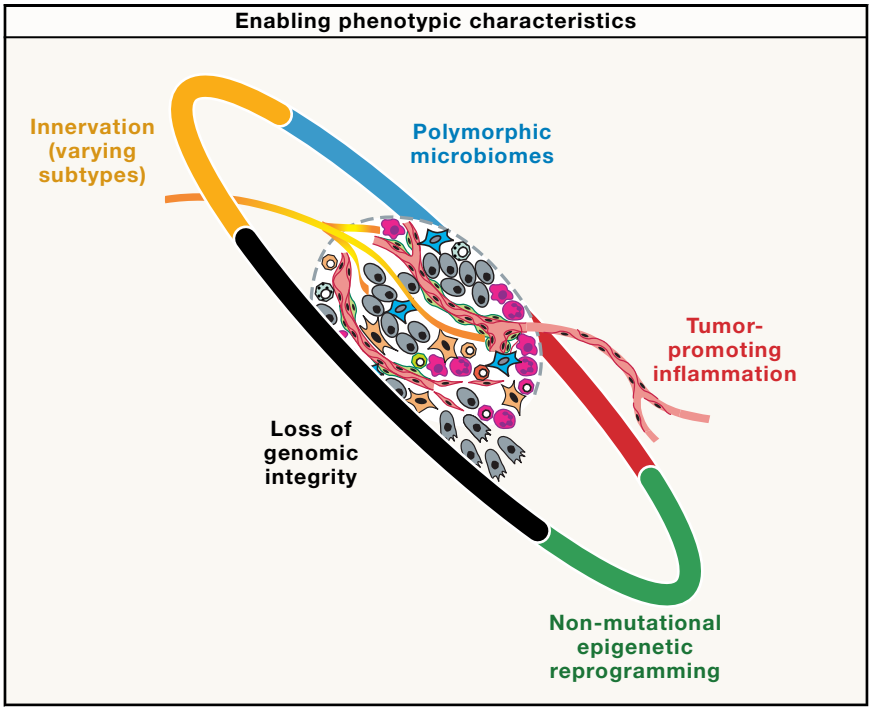


Figure 3. Enabling phenotypic characteristics

Although not functional mechanisms per se, five aberrant or misappropriated traits are instrumental in contributing to the acquisition of hallmark capabilities, being variously engaged in different tumor types and at distinctive stages of multistep tumorigenesis and subsequent tumor evolution.

phenotypic characteristics that facilitate their activation and selection during tumorigeneses and tumor progression. The most well-defined hallmark-enabling phenotypic characteristics are outlined below and summarized in Figure 3.

Loss of genomic integrity—Gene mutations and CIN

The foundation of cancer as a chronically expansive disease lies in the loss of genomic integrity. There are multiple manifestations of this phenotypic characteristic across the spectrum of cancer types, ranging from point mutations to amplifications, rearrangements, and deletions of genes and chromosomal domains; gains and losses of entire chromosomes (or arms); and the generation and persistence of linear and circular ecDNA, as well as whole genome doubling and genome shattering (chromothripsis). As noted above, telomere erosion and dysfunction prior to telomere stabilization by telomerase or ALT fuels this genomic disarray during tumorigenesis, facilitated by deficiencies in DNA damage-sensing and repair pathways, most prominently involving defects in mismatch repair and homologous recombination. The resulting disruptions in genome integrity elicit changes in gene expression and/or activity that convey hallmark capabilities.

The foundation of cancer as a genetic disease was laid by the serial discoveries that (1) DNA and RNA "tumor" viruses carry genes capable of transforming cultured cells and inducing tumors in experimental animals; (2) cellular genes, including proto-oncogene homologs of transforming genes transduced by RNA tumor viruses—e.g., *KRAS* and *MYC*—can be upregulated and/or mutated to do the same in the form of "driving oncogenes"; and (3) other genes are mutationally inactivated, including the aforementioned *TP53* tumor suppressor, as well as *RB*, *CDKN2B*, etc. Moreover, as noted above, expression-enhancing point mutations in the promoter for the gene encoding the TERT component of the telomerase enzyme are the most frequent non-coding mutations affecting hallmark capabilities in human tumors. Beyond these prominent culprits, more than 750 other genes are mutationally altered in human cancers and functionally validated as oncogenes or tumor suppressors (COSMIC) (Table S1).

For a plethora of other genes, changes in expression due to chromosomal aberrations—rearrangements that affect topological chromosomal domains, copy-number alterations, residence in ecDNAs, etc.—without identifiable gene-altering mutations are sufficient to promote hallmark capabilities. Indeed, CIN is evident in a majority of cancer types, which can be clonal, following episodic instability, or dynamically changing during tumor evolution. Aneuploidy/CIN in the absence of demonstrable oncogenic mutations has been associated with proliferation, evading growth suppression, plasticity, metabolism, and immune evasion, suggesting that complex patterns of dysregulated gene expression are also functionally important consequences of the loss of genomic integrity and consequent instability. Moreover, while function-altering point mutations in oncogenes and tumor suppressors are deeply embedded in our conceptualization of cancer genetics, the realization that normal cells throughout the body are replete with asymptomatic oncogenic mutations, e.g., in *KRAS* and *TP53*, among many, informs us that such mutations cannot fully explain the inception and progression of tumorigenesis, which evidently also involves most of the other hallmark-enabling phenotypic characteristics discussed below, notably tumor-promoting inflammation and innervation.[7,10,15,17,95–99]

Non-mutational epigenetic reprogramming

It is now evident that non-mutational reprogramming of gene expression across the genome is a separable enabling phenotypic characteristic, markedly distinct from mutational cancer genetics. Its existence is actually quite logical, given that most multicellular organisms develop from fertilized eggs via epigenetic programming events that differentiate daughter cells during embryogenesis and organogenesis, a process that continues during systemic homeostasis to maintain differentiated cell fates and functions. Beyond the genomic rearrangements that mediate development and antigenic specialization of B and T lymphocytes, these physiological processes are not based on

genetic alterations and, as such, are purely epigenetic. Similarly, there are increasing examples of non-mutational programming of regulatory circuits that convey hallmark capabilities in cancer. Indeed, a new field of cancer research has been spawned.

Certainly, there is crosstalk with mutational cancer genetics, involving function-modifying mutations in chromatin organizing and imprinting genes as well as rearrangements of chromosomal domains, which are facets of the aforementioned characteristic of hallmark-enabling genome instability, laying a foundation of regulatory states in cancer cells. Upon this foundation, the dynamic flexibility of epigenetic reprogramming underlays phenotypic plasticity, heterogeneity, and pathogenic evolution of cancer cells, e.g., in enabling metastatic capabilities. There are at least three layers to epigenetic reprogramming of hallmark-enabling gene expression: (1) changes in the accessibility (or inaccessibility) and structural conformations of the chromosomal domains in which such genes reside, mediated in part by DNA and histone modifications; (2) the availability and activity of TFs and multi-protein complexes that bind to gene regulatory elements so as to stimulate or repress transcription from accessible genes; and (3) post-transcriptional and post-translational modifications of mRNA and proteins, respectively.

Mechanistically, beyond mutations in epigenetic regulators, arguably a facet of the preceding parameter, there is increasing evidence for a variety of purely epigenetic mechanisms. These include feedforward and feedback signaling loops within cancer cells that sustain reprogrammed regulatory states, as well as paracrine signals conveyed both locally from the TME and systemically from the blood-transporting circulatory system and the nervous system. As proof of concept, some pediatric tumors are devoid of recurrent (driving) genetic abnormalities and are arguably governed by purely epigenetic mechanisms.

More broadly, while its importance in the scheme of enabling phenotypic characteristics is increasingly well established, there is much to be learned about mechanisms and manifestations. For most tumors, non-mutational reprogramming is likely to be insufficient and is rather intertwined with hallmark-enabling gene mutations and chromosomal rearrangements and instability. Notably, non-mutational epigenetic reprogramming is a major determinant of the capability for phenotypic plasticity discussed above, and their cooperative intersection evidently can contribute to all stages of multistep tumorigenesis and tumor evolution, including to states of therapeutic resistance.[3,64,73,79,100–104]

Tumor-promoting inflammation

While inflammation of tumors by activated T cells and subsets of innate immune cells can be tumor-destructive and hence beneficial, it is now well established that infiltration of neoplastic lesions and tumors by alternative subsets of most immune cell types collectively promotes tumorigenesis and subsequent tumor progression by conveying in paracrine fashion hallmark functions. Basically, the tissue abnormalities created by developing and progressing tumors induce and corrupt restorative and protective physiological mechanisms, of which the most prominent example involves wounded tissue. Macrophages and neutrophils are recruited to help orchestrate wound healing by contributing to the stimulation of epithelial cell proliferation and migration, angiogenesis, and limitation of adaptive immune responses (to avoid autoimmunity). Indeed, wound healing demonstrably involves five of the nine cancer hallmarks, namely sustaining proliferative signaling, resisting programmed cell death, inducing angiogenesis, evading immune destruction, and activating invasion and migration. Recruited innate immune cells are also part of the equation. Whereas these capabilities are transiently operative during wound healing, they are sustained in growing tumors, reflecting the historical observations of pathologists that tumors are wounds that fail to heal.

The roster of tumor-promoting inflammatory cells includes phenotypic states and subclasses of virtually every immune cell type, as will be discussed below in "Hallmark-conveying cells in the tumor microenvironment." Mechanistically, immune cells are recruited into and phenotypically programmed within incipient neoplasias, primary tumors, metastases, and tumor-draining lymph nodes in response to a panoply of tissue damage and abnormality/stress signals, including hypoxia, damage-associated molecular patterns (DAMPs), inflammatory forms of cell death, senescence, microbial infection, translocation across mucosal/epidermal barriers, etc. Signaling systems such as the abnormality-sensing Toll-like receptor (TLR) pathway, the DNA damage-sensing cyclic GMP-AMP synthase-stimulator of interferon genes (cGAS-STING) pathway, and the stress-sensing nuclear factor κB (NF-κB) pathway have context-dependent roles in the induction and propagation of inflammatory responses, alternatively being capable of orchestrating tumor-antagonizing T cell-mediated immunity or tumor-promoting and immunosuppressive innate immune infiltration. In most symptomatic tumors, the latter is evidently favored.

Oncogenic transformation can also directly recruit tumor-promoting immune cells by inducing the expression in cancer cells of pro-inflammatory cytokines and chemokines, in part through modulating the aforementioned signaling pathways. Thus, for example, mutant *KRAS* and mutant/inactivated *TP53* genes can each elicit distinctive signals capable of recruiting tumor-promoting myeloid cells. Conversely, precocious inflammation due to wounding and other tissue abnormalities has been implicated in the inception of tumorigenesis by inducing inappropriate proliferation of quiescent normal cells, including ones carrying preexisting latent oncogenic mutations, as well as others suffering new oncogenic mutations—e.g., from inflammatory reactive oxygen species (ROS)—or receiving paracrine proliferative signals from adjacent inflammatory cells. In addition, special subsets of innate and adaptive immune cells have potent immunosuppressive activities that curtail immunosurveillance, thereby favoring progression of pre-neoplastic lesions and frank tumor outgrowth.

Collectively, the evidence is compelling that the enabling phenotypic characteristic of tumor-promoting inflammation is instrumental across cancer types. It is evoked by diverse mechanisms that vary according to the cell and tissue of origin, by the stage of tumorigenesis and metastatic seeding, by the phenotypic states of "accessory cells" in their microenvironments, by the particular constitution of genetic and epigenetic states operative in the cancer cells, and by the characteristics of their genomic and cellular damage.[46,105–112]

Innervation

Pathologists have long recognized the relationship between tumors and nerves, particularly at tumor margins and along nerve sheaths (perineural invasion). Today, cancer neuroscience is experiencing a renaissance, propelled by advances in phenotypic and functional imaging, refined methods for tracing neuronal circuits and innervation, and increasingly precise genetic and pharmacological tools that enable gain- and loss-of-function perturbations of tumor-associated neurons and their signaling circuits across diverse cancer models, including optogenetic and chemogenetic approaches.

These convergences have substantiated a growing realization, namely that peri- and intra-tumoral innervation can modulate cancer phenotypes, including multiple hallmark capabilities. Both classes of peripheral nerves—autonomic, with sympathetic and parasympathetic branches, and somatosensory—and multiple types of neurons in the brain have, in different cancer contexts, been functionally implicated in facilitating hallmark capabilities via axonal proximity to cancer and accessory cells in the TME. Innervation has been linked to the capabilities for stimulating proliferative signaling, evading programmed cell death, mediating immune evasion, and enhancing invasion and metastasis, in large part via the secretion of a variety of neurotransmitters that signal by stimulating receptors (perplexingly) expressed in both cancer and accessory cells.

Remarkably, the communication from neurons is increasingly appreciated to involve more than simple physical proximity between axonal projections and cells in the TME. Varying forms of synaptic connectivity are being observed in primary brain tumors, in brain metastasis, and potentially in peripheral tumors and their metastases. Thus, in the CNS, glial cancer cells and a variety of metastatic cancer cells establish bona fide synapses with presynaptic neurons through which neurotransmitter signaling elicits regulatory electrochemical currents in the cancer cells. Additionally, brain-metastatic cancer cells can mimic astrocytes in pseudo-tripartite synapses to activate paracrine neurotransmitter signaling that stimulates their colonization as brain tumors. Functionally significant association and synaptic-like connectivity of axons and cancer cells have been described in pancreatic adenocarcinoma and primary and brain-metastatic small cell lung cancer, which could presage the involvement of neuron-to-cancer-cell-synaptic and pseudo-synaptic signaling in enabling hallmark capabilities for other peripheral tumors.

Additional forms of connectivity—both in CNS and peripheral tumors—involve the formation of adhesions and bridges involving neuronal cell-cell adhesion molecules that focus and intensify ligand-receptor signaling via intimate proximity, as well as tubular connections that enable transport of cytoplasmic modulators. Conversely, connected cancer cells can send reciprocal paracrine signals that induce neo-innervation and/or elicit proximal associations and programming of hallmark-enabling neuronal signaling. Neurons also release signals that recruit immunosuppressive, tumor-promoting inflammatory cells, mirroring their role in wound healing and underscoring the convergence of neural and immune programs that enable cancer hallmarks. Importantly, not all innervation is tumor-promoting, highlighting the complexity of neuronal interactions with tumors, albeit beyond the scope of this perspective.

Much like tumor vascularization, which involves a variable mix of angiogenesis and co-option of preexisting tissue vessels, tumor innervation evidently involves a blend of cancer-induced axonogenesis with invasive co-option and reprogramming of preexisting axons and their distal neuronal cell bodies. Notably, cancer cells can also express receptors for neuronal regulatory circuits that, via autocrine and paracrine (TME) ligand stimulation, similarly modulate hallmark capabilities in the absence of demonstrable neuronal connectivity. While additional functional contributions can be anticipated, it is already clear that innervation should be added to the roster of hallmark-enabling phenotypic characteristics and arguably considered as a potential hallmark capability in and of itself.[5,66,113–123]

Polymorphic microbiomes

It is now well recognized that the symbiotic relationship between mammalian organisms and diversely variable communities of bacteria and fungi, as well as viruses, archaea, and protozoa, which collectively constitute "microbiomes" populating different tissues, has profound modulatory effects on cancer pathogenesis and specifically hallmark parameters. The constitutions of microbial species can vary significantly from one individual to another and hence are polymorphic, most notably in regard to their effects on tumor evolution and responses to therapy, which can be alternatively antagonistic or protagonistic. While the gut microbiome is the prototype and proof of concept for the significance of polymorphic microbiomes in modulating cancer phenotypes, partially overlapping and demonstrably instrumental communities of microbes are localized in the skin, the lung, the uterine cervix, the urinary tract, and the oral cavity, among other sites. Although bacteria have been reported inside cancer cells and within the interstitial compartment of tumors, both their prevalence and functional effects on tumors are controversial and remain to be fully illuminated. Technologies to genetically profile the populations of microbiomes in all of these locations, as well as the capabilities to functionally perturb microbiomes with antibiotics and microbial transplantations in mouse cancer models and human cancer patients, have collectively revealed cancer-modifying effects and implicated particular species in their manifestation.

Through the lens of the hallmarks, microbiomes and particular species therein as well as their mis-localization—dysbiosis—have been shown to be capable of stimulating proliferative signaling and evading immune destruction, and effects on other hallmark capabilities can be anticipated but are currently opaque. Moreover, particular forms of microbial dysbiosis—including ones involving bacteria secreting mutagenic molecules—can demonstrably enhance the enabling phenotypic characteristics of genome instability and mutation and tumor-promoting inflammation, further highlighting the interplay between hallmark-facilitating parameters.

The most compelling evidence for the importance of polymorphic microbiomes centers upon immune evasion, in that microbiomes and particular species therein have been revealed to be capable of orchestrating immunosuppression and immune evasion, or alternatively stimulating anti-tumor immunity. Thus, fecal transplants from patients and mice that are responding to checkpoint-blocking immunotherapies into individuals that are

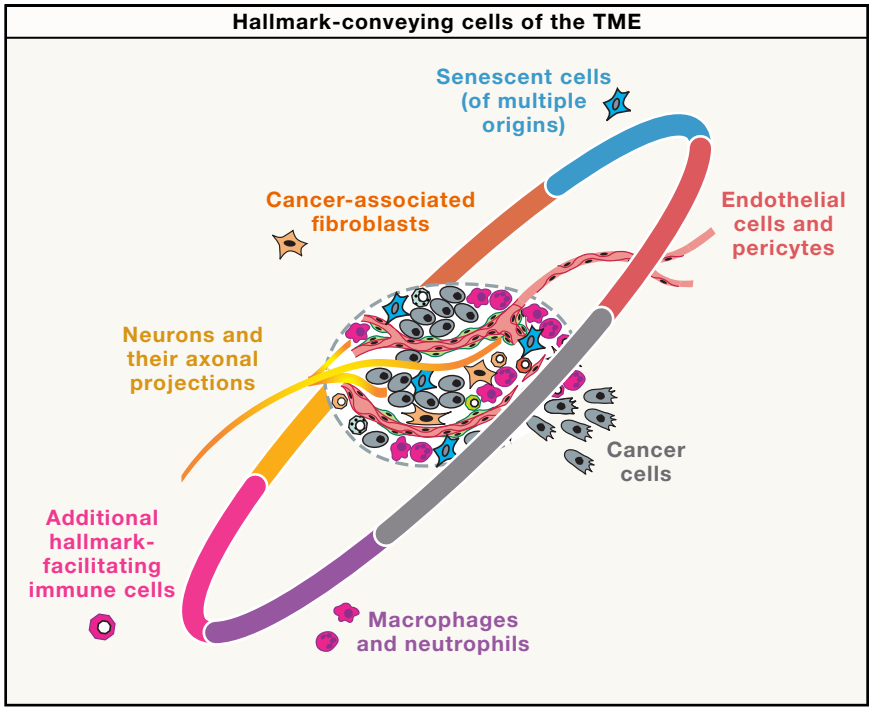


Figure 4. Hallmark-conveying cells of the TME

Tumors are cellular ecosystems, composed of cancer cells as well as a plethora of recruited and reprogrammed/corrupted heterotypic cells—"accessories to the crime"[4]—that populate neoplastic, primary tumor, and metastatic microenvironments. Notably, virtually all of the accessory cells can exist in multiple phenotypic substates, among which are ones that demonstrably contribute in different tumor contexts to most of the hallmark functional capabilities.

resistant can demonstrably disable or circumvent the functional capability to evade immune destruction, thereby enabling beneficial responses to such immunotherapies. Selective pharmacological ablation of tumor-promoting microbes or transplantation of tumor-antagonizing microbes can be similarly beneficial in mouse models and can be anticipated in cancer patients. It can be further envisaged that manipulation of tumor-promoting and -antagonizing microbiomes in the gut, and of the microbiomes residing in other tissues harboring proximal tumors, will also prove to be capable of disrupting additional, to-be-defined microbiome-facilitated hallmark functions, thereby improving responses to mechanism-targeted therapies beyond immunotherapy.[3,6,124–126]

HALLMARK-CONVEYING CELLS OF THE TUMOR MICROENVIRONMENT

In addition to the foundational presence of cancer cells, tumorigenesis and subsequent tumor progression involve a roster of recruited and/or locally reprogrammed normal cells—"accessories to the crime"—that are critical for the manifestation of cancer, which largely operates in the form of outlaw organs (Figure 4). Key features of the assemblage of cell types and substates typically associated with conveying the hallmark capabilities of solid tumors, as well as their progenitor lesions and metastases, are summarized below. Remarkably, substates of virtually all of the cell types described below, in addition to the cancer cells themselves, contribute to the capability to evade immune destruction.[4,46,81,127]

Cancer cells

Cancer cells, the genetically and epigenetically transformed derivatives originating from normal cells throughout the body, constitute the basis of this chronically expansive disease. As discussed above, cancer cells almost invariably contain function-altering mutations or chromosomal rearrangements in genes that via their aberrant expression and activity convey hallmark capabilities. Beyond a dozen or so frequently implicated oncogenes and tumor suppressors—*KRAS*, *MYC*, *EGFR*, *HER2*, *TP53*, *APC*, *RB*, and a suite of cyclin-dependent kinases and their inhibitors—over 750 genes that are genetically altered in tumors have been shown to contribute functionally to cancer cell intrinsic or tumoral phenotypes in some context. Of these, ~60% have been associated with hallmarks of cancer (Table S1). Remarkably, mutant *KRAS* has been variously associated in different tumor types with all of the nine hallmark capabilities, including cancer cell plasticity, as well as with the enabling characteristic of tumor-promoting inflammation. Importantly, however, expression of mutationally activated *KRAS* is evidently not sufficient, given multi-step tumorigenesis involving TSG and other mutations as well as epigenetic factors in *KRAS*-driven pancreatic and other cancers, and its asymptomatic presence in normal cells throughout the body.

Beyond genetic alterations, epigenetic reprogramming of cancer cells can influence virtually all of the hallmarks. Notably, most cancers contain heterogeneous populations of cancer cells that differ either genetically or purely epigenetically. Epigenetic variability can demonstrably involve feedforward auto-stimulatory signaling or dynamic variability in reaction to differing microenvironmental influences, in particular in response to a variety of stressful genomic and physiological conditions, empowered by the hallmark capability of phenotypic plasticity, as discussed above.

Another aspect of heterogeneity and plasticity involves reservoirs of CSCs, which like normal tissue stem cells often reside in special niches, proliferate episodically, and are less visible to the immune system, with the potential to spawn expansively proliferative cancer cells, which in some contexts can reciprocally spawn new CSCs that preserve their special contributions to the disease. CSCs have been defined historically by their enhanced efficiency at seeding tumors upon transplantation of cancer cells fractionated from tumors or populations of cultured cancer cells. This and other assays have revealed dramatic

variations in the frequencies of CSCs in different tumor types, which is not well understood. Infrequently proliferating or dormant CSCs residing in protective niches have been implicated in evasive resistance and relapse to many forms of cancer therapy in a variety of tumor types, highlighting their importance. However, there is increasing evidence that cancer cells can harbor a spectrum of stem-like qualities—"cancer stemness"—in a continuum of cell states that motivates a conceptual refinement of the CSC hypothesis. Another unrelated, albeit intriguing, observation is that cancer cells of multiple origins, most without any neuronal heritage, can upregulate one or more of a roster of receptors for neurotransmitters and neuronal growth factors that enable both paracrine signaling (from innervation) and autocrine stimulation of pathways that convey hallmark capabilities.[5,10,15,29,64,67,128–132]

Heterogeneity, plasticity, and non-mutational epigenetic reprogramming extend beyond cancer cells to virtually all of the accessory cells recruited to populate neoplastic, primary tumor, and metastatic microenvironments, as elaborated below.

Endothelial cells and pericytes

Neoplastic lesions, tumors, and metastases have a vasculature that is composed of tube-forming endothelial cells and perivascular supporting cells (pericytes), consequent to the induction of angiogenesis as well as the co-option of preexisting quasi-normal tissue vessels. Encompassing these two classes of tumor vasculature are multiple endothelial cell states, revealed by morphology and, increasingly, by single-cell and spatial transcriptomics analyses, collectively revealing considerable heterogeneity and evident plasticity. The endothelium of the angiogenic neovasculature is composed of tip, stalk, and mature tube-forming luminal cells, with the latter also constituting co-opted arterioles and venules that normally connect to capillaries. Additionally, other specialized endothelial cell states are involved in (1) the function of high endothelial venules (HEVs) that serve as portals for lymphocyte extravasation; (2) the formation of peri- and intra-tumoral tertiary lymphoid structures (TLSs) that assemble T and B cells in the context of anti-tumor immune responses; and (3) transitions from endothelial cells into mesenchymal cell states (EndMTs) that are in turn programmed to become CAFs.

Pericytes of the tumor vasculature also evidence tumor-enhancing phenotypes distinctive from their quiescent normal counterparts. Focusing on contributions of the tumor endothelial cells and pericytes to hallmark parameters, it is now well established that these cell types contribute to stimulating proliferative signaling, circumventing programmed cell death, evading immune destruction, fueling metabolism, and facilitating invasion and metastasis. The supply of oxygen, serum growth factors, and nutrients is but one component. Growth and regulatory factors are supplied in paracrine fashion to other cells within the TME by heterogeneous endothelial cells and pericytes in different vascular sub-compartments and phenotypic states, governed by the local milieu of microenvironmental signals within the tumor microenvironment.

The angiogenic tumor vasculature is typically immunosuppressive for T cells, involving multiple mechanisms, including repression of lymphocyte homing receptors needed for efficient extravasation, expression of ligands that induce T cell paralysis or death, and impaired formation of HEVs and TLS. Notably, both endothelial cells and pericytes contribute to the maintenance of an immunosuppressive TME. Another factor is the leakiness of the angiogenic vasculature, consequent to VEGFA-dependent and -independent signaling, which can enable cancer cells to intravasate into and disseminate through the blood and lymphatic circulatory systems to seed distant metastases. In parallel, tumors can release regulatory factors into the circulation that reprogram distant tissue vasculature to facilitate adhesion and extravasation of cancer cells with the potential to seed metastatic growths.[45–47,49,53,54,82,133–137]

Looking beyond cancer cells and their supporting vasculature, all of the discrete cell types described below can adopt phenotypes that either promote or suppress tumor growth. The list includes both quasi-stable subtypes as well as more dynamic cell states programmed by the local milieu in which they reside. The focus here will be on those states contributing to hallmark capabilities, leaving the description of alternatively passive and antagonistic states to other reviews.

CAFs

Virtually all tumors contain populations of fibroblastic cells that are (1) functionally abnormal and (2) diversely heterogeneous in cellular and molecular characteristics, involving dynamic plasticity in response to local microenvironmental stimuli. CAFs can be variously derived from co-opted and reprogrammed resident tissue fibroblasts, from bone-marrow-derived mesenchymal stem cells, and via lineage plasticity involving *trans*-differentiation from endothelial cells, pericytes, and in certain tissues undergoing tumorigenesis, adipocytes, astrocytes, and vascular smooth muscle cells. Transcriptomic and proteomic profiling and functional genetic studies have identified several common, quasi-stable phenotypic classes, along with dozens of sub-states and hybrids thereof. The most common classes have been dubbed myofibroblastic (myCAFs) and inflammatory (iCAFs) by function and molecular profiling. As with most cell types in the TME, CAFs can have dichotomous effects on tumors, ranging from being tumor-suppressing and pro-inflammatory for T cells to tumor promoting and immunosuppressive.

Through the lens of contributing to hallmarks, CAFs have been documented to have the capabilities to stimulate proliferative signaling in cancer cells, to inhibit programmed cell death, and to induce and modify tumor vasculature, all by releasing proteins and other regulatory factors, including exosomes. CAFs can also remodel the extracellular matrix, thereby modulating mechanotransduction of regulatory signals, and they can relieve the paracrine growth suppression that is operative in some normal tissue fibroblasts. Moreover, CAFs can fuel cancer cell energetics via metabolic symbiosis, suppress adaptive immunity to evade immune destruction, and support invasion and metastasis, as well as recruit pro-angiogenic, immunosuppressive myeloid cells. Importantly, these diverse functional contributions are not similarly operative in all tumor types or stages of tumorigenesis and progression but are context dependent in regard to the populations of CAF states and sub-states, their abundance, and their localization in the TME.[63,138–144]

Macrophages and neutrophils

The aforementioned enabling characteristic of tumor-promoting inflammation principally involves phenotypic states of TAMs and TANs, which are recruited as immature myeloid cells from the bone marrow and subsequently programmed into distinct phenotypic states en route to and inside the TME. Additionally, tissue-resident macrophages can be co-opted—much like resident fibroblasts—and reprogrammed. Molecular profiling and functional perturbations have revealed that TAMs and TANs can adopt phenotypic states spanning from tumor-antagonizing to diversely tumor-promoting, including phenotypic mixtures thereof.

Macrophages were initially framed in terms of M1- and M2-like states, being phagocytotic and antigen-presenting versus alternatively activated, respectively, a concept that helped organize early observations of their pro- and anti-inflammatory effects. It is now appreciated that the convenient designations of M1/M2-like TAMs represent phenotypic extremes in a spectrum of substates that convey distinctive functional roles in tumors. An analogous spectrum from tumor-antagonizing to tumor-promoting phenotypic substates is evident for TANs. Moreover, sub-states of TANs and TAMs, in particular their immature progenitors, are also described as granulocytic and monocytic myeloid-derived (immuno-)suppressor cells (MDSCs), respectively, reflecting their contributions to this instrumental hallmark capability. Focusing on their well-defined functional contributions to cancer, hallmark-conveying substates of macrophages and neutrophils can demonstrably, via multiple mechanisms, stimulate cancer cell proliferation and angiogenesis, suppress adaptive immunity, and facilitate invasion and metastasis. While both TAMs and TANs are largely post-mitotic, with variably brief half-lives, such that new cells are chronically immigrating into the TME to replace those that have died, there is intriguing evidence for self-renewing and/or atypically long-lived (and senescent) TAMs and TANs, including reprogrammed tissue-resident TAMs, which may help sustain their hallmark-supporting contributions to tumor progression. Importantly, substates of TAMs and TANs can demonstrably convey hallmark-enabling capabilities to all stages of multistep tumorigenesis, as well as to tumor evolution, including to states of adaptive resistance to therapy, albeit with considerable variability in different cancer types, including the aforementioned substates that have hallmark-opposing effects.[46,49,145–153]

Additional hallmark-facilitating immune cells

Reflecting the complex regulation of innate and adaptive immunity, most other cell types of the immune system also can adopt immunosuppressive and variously hallmark-conveying phenotypic states that contribute to cancer. Dendritic cells, which normally serve to activate CD8 and CD4 T cells, can be pacified or rendered dysfunctional by paracrine signals in the TME. NK cells, which should attack non-immunogenic (major histocompatibility complex [MHC]-low) cancer cells, can assume states in tumors that are passive and/or even immunosuppressive for T cells, as can other innate lymphoid cells (ILCs). Moreover, there is evidence that basophils, mast cells, and eosinophils can be programmed in the TME to adopt phenotypic cell states that are stimulatory for multiple hallmark capabilities.

The compendium also includes Treg and Breg that normally participate in switching off adaptive immune responses to limit the potential for chronic T/B cell activity and consequent autoimmunity. Most prominent are Tregs, whose recruitment and programming in tumors can demonstrably contribute to immune evasion by pacifying activated, cytotoxic CD8 and CD4 T cells. Tregs are an interconvertible substate of conventional CD4 T cells. Tregs are recruited and programmed by cytokine-induced expression of the FOXP3 TF, shifting their identity from immunostimulatory helper and cytotoxic CD4 T cells to regulatory cells that suppress—using a multiplicity of mechanisms—expansion and effector functions of CD8 and CD4 T cells and dendritic cells. Moreover, Tregs can have broader effects on hallmark capabilities, including secretion of VEGFA to stimulate angiogenesis, etc., ostensibly reflecting roles of Tregs in wound healing and tissue regeneration that are repurposed by tumors.[46,154–162]

Neurons and their axonal projections

As discussed above, it is increasingly evident that most tumors are innervated by a variety of neuronal types and subtypes, both within the CNS as well as by sympathetic, parasympathetic, and somatosensory neurons of the peripheral nervous system that innervate the body. While the nuclei of these cells are most commonly located in distal ganglia and not inside tumors per se, neurons, and in particular their axonal projections into tumors, can now be considered as generic constituents of the TME.

There is much to be learned about the diverse consequences of tumor-associated neurons, both locally and systemically, spanning from tumor-promoting to tumor-antagonizing to other physiological effects. Their interactions involve reciprocal programming of phenotypic states both in cancer cells and neurons, as well as in many of the heterotypic cell types that constitute the TME. The hallmark-facilitating roles of neurons are evident in a growing spectrum of cancer types, involving paracrine signaling interactions of axons with proximal cancer cells and other cell types in the TME. Bona fide synaptic transmission of electrochemical signals as well as pseudo-synaptic connections conveying qualitatively distinct signals collectively program functional capabilities and phenotypic states in cancer cells and TME cells, directly and indirectly.

Neurons and their projections have been shown in different tumor contexts to stimulate proliferative signaling and angiogenesis, to inhibit programmed cell death, to suppress adaptive immunity, and to foster invasion and metastasis, along with recruiting tumor-promoting immune cells, with all of these effects in part likely reflecting the roles of innervation during wound healing, including of wounded nerves. Moreover, a growing number of case studies have variously demonstrated that neuronal projections can program cancer cell plasticity, heterogeneity, and function, as well as phenotypic states in endothelial cells and pericytes, CAFs, TAMs and TANs, T cells, and dendritic cells, all to the benefit of tumor progression. As such, there is now a compelling rationale to add neurons and their projections to the canonical roster of hallmark-conveying cell types in the TME.[5,113,117–122]

Senescent cells (of multiple origins)

It is well established that senescent cells—having transiently or irrevocably lost proliferative potential in neoplasias, tumors, and metastases—can functionally modulate tumorigenesis and tumor progression in large part through induction of a heterogeneous senescence-activated secretory program (SASP) composed of signaling proteins and lipids, exosomes, etc. Senescent cancer cells have historically been characterized as inherently tumor-suppressing due to their abrogated capability for cell proliferation, variously induced by hyperactivated oncogene-induced senescence pathways, by insurmountable DNA damage and cell stress, by aging "cells of origin," as well as by SASP signaling that elicits their clearance by immune cells. However, senescent cancer cells and an increasing assemblage of senescent accessory cells in the TME have been documented to be programmable into tumor-promoting variants via changes in their SASP. The mechanisms underlying the heterogeneous regulation and dichotomous effects of the SASP on tumor phenotypes, even among cancer cells from the same tumor type, remain a mystery.

Our clearest understanding about the effects of tumor-promoting senescence involves cancer cells. In certain contexts, senescent cancer cells can demonstrably simulate, via their SASP, proliferative signaling in their nearby non-senescent brethren, paracrine protection against programmed cell death, angiogenesis. suppression of adaptive immunity, and recruitment of innate immune cells with similar capabilities. CAFs and aging normal fibroblasts, which can develop CAF-like phenotypes, can also senesce into variants that support initiation as well as progression of tumorigenesis, again via stimulating hallmark capabilities. Case studies also implicate senescent tumor endothelial cells and senescent TAMs and TANs in fostering tumor progression. Interestingly, aging macrophages tend to display M2-like tumor-promoting characteristics, which persist upon becoming senescent. Since macrophages are short-lived, it seems plausible that senescent TAMs, being persistent, might help sustain the effects of their viable but transitory counterparts. A challenge for the future will be to clarify the differences between post-mitotic but long-lived TAMs and TANs from demonstrably senescent TAMs and TANs in regard to their hallmark-promoting capabilities and their persistence times in tumors, and hence their respective impact on tumorigenesis and subsequent tumor evolution.

Finally, case studies involving pharmacological ablation of senescent cells of various origins support the importance of their tumor-promoting functionality and evident complementarity to the contributions of their viable compatriots, suggesting that strategies to therapeutically target viable cells in the TME should include consideration of co-targeting their senescent derivatives. Therapeutic targeting of senescent cells will, however, require deeper knowledge about the diversity of hallmark-impinging senescent cell phenotypes that are evident in different (aging) tissues and tumor types, stages of tumorigenesis and tumor progression, and between individuals with the same disease state.[3,163–170]

In addition to this consortium of heterotypic cells that variously contribute to hallmark capabilities, the physical microenvironment of solid tumors regulates cell phenotypes and the tumor-promoting mechanisms they convey. Modulatory characteristics include the composition of the extracellular matrix and of the basement membranes defining tissue boundaries and cell compartments, as well as matrix stiffness, interstitial fluid flow, acidity, and hypoxia.[46,100,171,172]

INTERACTIONS OF CANCER AS A SYSTEMIC DISEASE

The hallmarks of cancer perspectives have largely focused on the concept of tumors as "outlaw organs," finessing the fact that tumors are connected to and interactive with the body in the form of a systemic disease, including but not limited to the functional contributions of the impinging circulatory and nervous systems and polymorphic microbiomes. A plethora of parameters—dubbed "the clouds of complexity"—overlay the simplicity of the hallmarks and motivate illumination of their modulatory roles for mechanisms of tumor development and evolution and for their potential in furthering the strategic goal of cancer research to improve therapeutic efficacy and quality of life for cancer patients.[6] By way of illustration, two such modulatory complexities are highlighted due to their demonstrable hallmark-enabling characteristics: aging and obesity.

Aging

The basis for the long-established correlation between aging and cancer incidence is multifactorial and exceedingly complex. Nevertheless, it is evident that aging affects many of the parameters discussed above. Aging tissues accumulate cells carrying latent, asymptomatic oncogenic mutations and other losses of genomic integrity. Moreover, ostensibly normal cells within aging tissues—putative cells of origin—can become more plastic and less well-differentiated, and consequently more prone to facilitating tumorigenesis. As noted above, while age-related increases in the abundance of senescent cells may not spawn incipient neoplasias per se, their SASPs can stimulate paracrine proliferative signaling and innate immune cell inflammation with the potential to trigger tumorigenesis in neighboring non-senescent cells. It is important to note that, in addition to the paracrine effects of senescent cells, age-related secretory changes in non-senescent aged cells can also impact tumorigenesis, ranging from alterations in the biophysical structures (e.g., extracellular matrix [ECM]) in which incipient cancer cells reside, to induction of angiogenesis, and shifts in bioavailable metabolic supplies and their utilization.

Additionally, alterations in the structure, cellular composition, and functionality of aging tissues can promote selection for "normal" cells containing mutations that provide adaptive advantage for their persistence in the aged context. One potential consequence is that some mutant clones may have concordantly acquired hallmark capabilities that facilitate tumorigenesis. For example, the aging hematopoietic system can develop clonal expansions and biases in differentiation that favor the overproduction of immunosuppressive, hallmark-promoting macrophages and neutrophils. Additionally, the fitness and diversity of the adaptive immune system decline with age, and its activity is further suppressed by the appearance of immunomodulatory CAFs, TANs, and TAMs, and potentially their persistent senescent derivatives, in developing neoplasias and tumors.

Long-term exposure to environmental agents—"tumor promoters"—that further stimulate wound-healing-like inflammatory responses can induce and/or amplify many of these factors. Indeed, in addition to chronic inflammation, there are many overlapping "hallmarks of aging," including losses in genomic integrity, epigenetic reprogramming with resultant phenotypic plasticity, and pathologic imbalances in microbiomes. The concomitant advances in knowledge and the intersection of the fields of aging and cancer hold promise for distilling this complexity into a deeper understanding of underlying mechanisms with prospects for impactful intervention in the inception and progression of cancer.[6,164,166,173–177]

Obesity

Obesity is rising steadily in the populace worldwide, and there is an increasingly compelling epidemiological link between obesity and cancer risk. Similar to aging, the mechanistic effects of obesity on tumorigenesis and tumor progression are also very complex, multifactorial, and variable across the spectrum of human cancers. Through the lens of the hallmarks, there are emerging associations that bear mention, with the precaution that much remains to be learned about the systemic effects of obesity on cancer.

First and foremost are adipocytes, fat-storing cells that markedly expand in numbers and fat content in obese tissues. Adipocytes are in addition endocrine cells with diverse modulatory roles, centered in part on the differential secretion of two hormones—leptin and adiponectin—along with a plethora of other hormones and growth factors. Adiponectin is preferentially secreted by lean adipose tissue, whereas leptin is preferentially secreted by obese adipocytes, wherein adiponectin secretion is reduced. This latter hormonal imbalance is associated with tumor-promoting actions-at-a-distance on incipient and developing neoplasias, tumors, and metastases, as well as paracrine effects when adipocytes are constituents of the TME. As such, obesity has the potential to increase cancer risk as well as to enhance tumor evolution.

Notably, the obesity-dependent endocrine effects on the immune system are dichotomous and context dependent, ranging from immunosuppression to promoting anti-tumor immunity. Currently, heightened levels of leptin and other (obese) adipocyte-derived regulatory factors are most clearly implicated in sustaining cancer cell proliferation, inducing angiogenesis, stimulating chronic tumor-promoting inflammation, evading—in some contexts—immune destruction, and altering tumor metabolism, with the latter additionally influenced by the hyperglycemia associated with obesity. There are experimental clues associating obesity with other hallmark capabilities and enabling phenotypic characteristics, and the tools are at hand to illuminate the mechanistic links between obesity and cancer, clarifying the prospects for their disruption by lifestyle changes, including diet and exercise.[6,178–180]

Other factors

Beyond these two illustrative examples, there are multiple facets to the complex interplay between cancerous lesions and bodily systems, alluded to in Figure 2 without subsequent delineation, which have been recently reviewed.[6] As discussed above, innervation, vascularization, and inflammation intrinsically involve systemic connectivity that affects many hallmark characteristics, as well as the reciprocal induction by tumors of pathogenic systemic abnormalities, including, for example, cachexia. Tumor-induced dysbiosis of the microbiome also has bidirectional effects. Additionally, systemic exposures to a diversity of environmental agents—smoking, air and water pollution, microplastics, etc.—disrupt normal organ systems and foster, directly and indirectly, hallmark functions in developing neoplasias and evolving tumors.[6,95,181,182]

FROM THEN TO NOW, LOOKING TO THE FUTURE

26 years on, the endurance, resonance, and continuing refinement of the hallmarks of cancer conceptualization raise a salient question: what does the future hold? Of course, other widely operative hallmark parameters may be revealed. Meanwhile, beyond illuminating aging, obesity, and other systemic modulators, there are tangible opportunities to further refine and delineate the current concept so as to advance knowledge about mechanisms of the disease and also to apply it to patient care. Two such goals are briefly outlined below.

Multi-omic profiling and non-invasive profiling and imaging of hallmark heterogeneity in human tumors

It was inherent in the original publication in 2000 and clearly evident in 2026 that there is widespread variability in the emergence of the hallmark-associated parameters during initiation and stepwise progression of multistage tumorigenesis and subsequent tumor evolution to states of heightened malignancy. This temporal variability manifests in the induction of hallmark capabilities, in the presence and contributions of hallmark-enabling phenotypic characteristics, and in the recruitment and programming of hallmark-facilitating cells in the TME (Figures 1, 3, and 4). Thus, depending on the barriers to and instigators of initiation and stepwise progression, it is logical that there are distinct temporal roles and engagements in different tumor types for instrumental components of all four of the parametric dimensions outlined in Figure 2.

Histopathological heterogeneity has long been recognized in premalignant lesions, tumors, and metastases, as revealed using histochemical and immunostaining to reveal tumor landscapes and TME cell types and states. Now, the development of powerful technologies for profiling cells and tissues is allowing roadmaps to be charted in previously unimaginable detail. Applicable technologies include single-cell RNA, DNA, and protein sequencing, as well as spatial transcriptomics and proteomics (and soon spatial genomics, including histone modifications and DNA methylation).

An encouraging example comes from transcriptome profiling of human tumors using gene signatures partially reflective of particular hallmarks, which revealed both regional heterogeneity and variability in the presence of hallmark-associated signatures in different human tumors.[183] This study encourages the possibility of more broadly and definitively applying single-cell and spatial tissue profiling technologies to illuminate molecular features of the cell types and substates, their importance for hallmark parameters, and their implicated contributions at different

stages in tumor development and evolution. Evaluation and application of this avalanche of multi-omic big data will be enabled by remarkable advances in bioinformatic and data-science algorithms involving machine learning and artificial intelligence, as exemplified by the publications cited below. These new technologies will serve to integrate and help elucidate the results forthcoming in the new frontier of "mechanistic digital pathology," in the present context by describing the presence and predicted importance of particular hallmark parameters for prognosis and therapeutic responsiveness. Unsupervised training of neural networks using large datasets of human tumors to create "pathology foundation models" is in the pipeline, with the potential to empower the illumination of hallmark features in tumor tissues.

While these groundbreaking technologies for mechanistic digital pathology will be highly informative, they rely on tumor biopsies or resections, which are typically limited by practical feasibility and the ability to perform serial sampling. As a result, they provide static temporal snapshots into the particular disease state, which may not accurately foretell the future of disease progression or actual responses to therapies. One solution may come from the rapidly evolving technologies for profiling tumor-derived molecules, organelles, and cells in the circulation—so-called "liquid biopsies"—that have yet to be applied to illuminating hallmarks per se but which can be envisaged to be applicable. Another will likely come from a complementary technology, also evolving in its capabilities, namely non-invasive imaging of hallmark-associated features. Indeed, some but not all of the hallmark parameters are demonstrably amenable to different non-invasive imaging modalities, and further innovation and applications can be anticipated.[183–194]

Prospects for hallmark-guided therapeutic co-targeting

While the hallmarks of cancer conceptualization offer heuristic value for the cancer research community and for students and others trying to understand the biological rationale for the daunting complexities of cancers, a corollary is whether the concept has any value for cancer medicine. In 2011, we introduced the possibility of therapeutic targeting of hallmark capabilities with mechanism-guided drugs, using illustrative examples of then-current drugs aimed at the eight hallmarks and two enabling phenotypic capabilities.[2] Fifteen years later, there are clinically approved drugs targeting proliferative signaling, abrogated tumor suppression, apoptosis, angiogenesis, and immune evasion. However, for most patients, adaptive resistance and relapse eventually develop to such hallmark-targeting monotherapies. Moreover, there are currently no clinically approved drugs, or even compelling preclinical candidates, for inhibiting telomerase or ALT, for disrupting metabolic diversity, for inhibiting invasion and metastasis, or for shutting off phenotypic plasticity. Hopefully, efficacious drugs will be developed in the coming decade, but even then, they face the reality of adaptive resistance.

One potential solution to the limitations of hallmark-targeting monotherapies is a strategy of hallmark co-targeting, namely treating patients with drugs that disrupt different and complementary functional capabilities, enabling phenotypic, characteristics, and accessory cells in the TME. Notably, co-targeting with distinctive drugs is a cornerstone of modern cancer therapeutics, often (but not exclusively) centered around the concept of hitting complementary vulnerabilities in cancer cells identified using *ex vivo* drug screening assays. Extending upon this clinically validated foundation, the current proposition is that attacking cancer "by land, by air, and by sea" will present challenging obstacles to the concomitant development of adaptive resistance to multiply targeted mechanisms and vulnerabilities, each involving a distinctive hallmark parameter.[195]

Three examples of the logic of hallmark co-targeting are presented by way of illustration. Two, which have been tested clinically and approved for treating certain cancer patients, constituting proof of concept, involve targeting the angiogenic tumor vasculature with VEGF/VEGFR inhibitors, which not only inhibit angiogenesis but also quasi-normalize the structurally and functionally aberrant vasculature, improving blood flow and facilitating the extravasation of T cells in the context of anti-tumor immune responses. One involves co-targeting angiogenesis with VEGF/VEGFR inhibitors and the immuno-evasive hallmark capability with ICIs (e.g., anti-PD1/anti-PD-L1) (Figure 5A). The rationale for this strategy is further bolstered by the fact that VEGF is known to cause immunosuppressive effects in the TME via multiple mechanisms. A number of such combinations have shown better efficacy than monotherapy and current standards of care in a series of late phase clinical trials of patients with renal, lung, hepatocellular, and endometrial cancers, leading to clinical approvals. However, similar trials with different regimens or different drugs from each class in these and other cancer types failed to demonstrate sufficient benefit for clinical approval, indicating the need for further refinement in trial designs. An example of such refinement is the promising development of bispecific monoclonal antibodies (mAbs)—ideal drugs, in principle, for hallmark co-targeting—that concomitantly block VEGFA and PD1 (or PD-L1). Bispecific anti-VEGFA/PD1 mAbs are showing demonstrable efficacy in late-stage clinical trials, though none have yet attained FDA approval, and widely varying responses are being observed in different tumor types.

The second example involves an anti-VEGFA mAb (bevacizumab) in combination with an inhibitor (olaparib) of the poly (ADP-ribose) polymerase (PARP) DNA repair enzyme, which targets a vulnerability presented by the enabling characteristic of genome instability consequent to mutationally impaired DNA damage repair pathways. This dual therapy has shown added benefit in certain patients with ovarian cancer, leading to clinical approval (Figure 5B). Another clinically approved example, in certain patients with acute myeloid leukemia (not shown), involves co-targeting resistance to cell death with the BCL-2 inhibitor venetoclax along with epigenetic reprogramming using azacytidine or decitabine, which inhibit DNA methylation and thereby disrupt hallmark-enabling gene expression.

The third example, which is intended to illustrate the logic of formulating new conceptual designs, ones that have yet to be clinically validated, involves the lack of generality for immunotherapy of solid tumors. While ICIs have demonstrated historic efficacy in patients with select tumors and subsets thereof, many tumor types do not show responses to monotherapy, or rather, respond transiently. As discussed above, one prominent mechanistic basis for intrinsic and adaptive resistance to ICIs and other

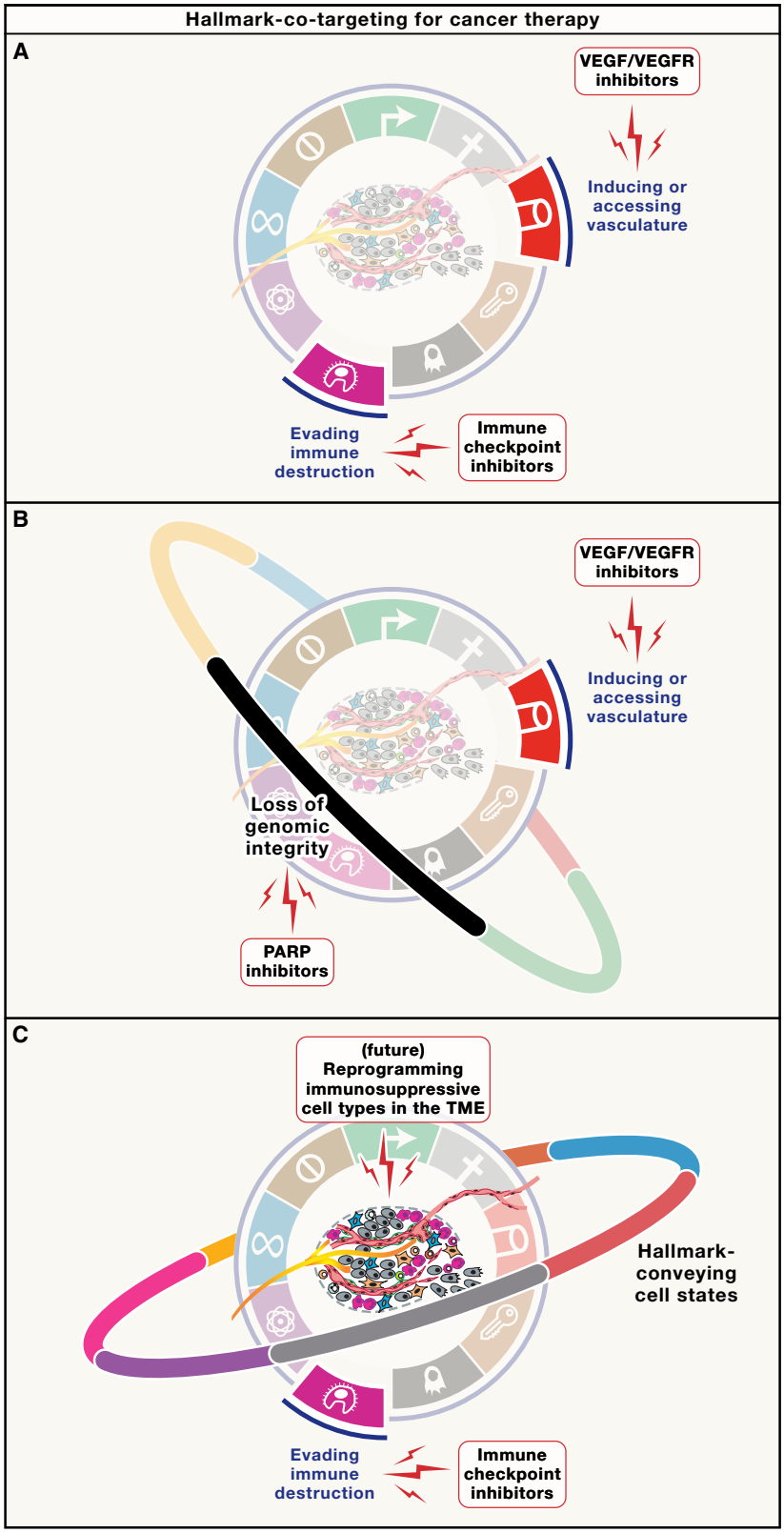


Figure 5. A rationale for therapeutic co-targeting of cancer hallmarks

It is increasingly evident that future improvements in cancer therapy will come in large part from increasingly sophisticated drug combinations, aiming both to improve efficacy and to circumvent intrinsic and adaptive resistance mechanisms to most monotherapies. The concepts summarized in this review suggest a therapeutic strategy involving co-targeting different cancer hallmarks, as well as hallmark-associated parameters in the other conceptual dimensions, with mechanism-guided drugs. It can be envisaged that eliciting concomitant adaptive resistance to mechanistically independent drug targets will prove to be challenging for co-targeted cancers, thereby prolonging efficacy and delaying relapse.

(A and B) Examples of clinically approved hallmark therapeutic co-targeting—"signs of light"—that encourage further evaluation of this logic.

(C) An as yet unrealized example of a therapeutic strategy involving ICIs combined with drugs disrupting effector mechanisms (or cells) operative in the immunosuppressive tumor microenvironment.

immunotherapies involves a multiplicity of cells in the TME that elaborate potent immunosuppressive activities. Therefore, a logical strategy is to block immune checkpoints on T cells concomitant with pharmacological reprogramming of CAFs, TAMs, TANs, cancer cells, and other accessory cells, or subsets thereof, to obviate their immunosuppressive capabilities (Figure 5C).

While the first two examples are demonstrably beneficial, albeit not transformative in terms of enduring survival benefit, the results encourage further investigation of hallmark co-targeting trial designs involving logical combinations of these and other cancer drugs—e.g., KRAS inhibitors—that disrupt different, quasi-independent hallmark parameters. Future hallmark co-targeting strategies may prove to have transformative value for cancer patients, not only in improved therapeutic benefits but also in their endurance, by raising the bar for relapse and progression, which would require the induction of multi-faceted adaptive resistance mechanisms for each of the distinctive mechanisms being targeted.[48,186,196–205]

IN CONCLUSION

The conceptual premise set forward in January of the year 2000 has endured, remarkably and unexpectedly, for 26 years, namely that there is a logical underpinning to the complexity and diversity of human cancer. From this foundation, refinements and embellishments have been put forward in a series of publications in the last decade and the current one that have expanded the landscape, albeit complicating the elegant simplicity of the original synthesis. This synopsis has sought to integrate many but not all of the various facets into a new framework, one that hopefully restores some of the conceptual clarity. By considering distinctive parametric classes as different dimensions (Figure 2), the various topical parameters are organized and distilled to crystallize in essence their individual and interactive contributions to cancer. Thus, here and now, there are nine acquired functional capabilities, the hallmarks of cancer, five enabling phenotypic characteristics, and seven classes of hallmark-facilitating cells in the tumor microenvironment, along with a number of instrumental systemic interactions, of which two hallmark-influencing examples are presented.

Looking beyond the here and now, to the future, two clear opportunities are discussed, namely (1) to elucidate and illuminate the involvement of the various hallmark parameters in different cancers throughout multi-step tumor development and tumor evolution to states of heightened malignancy and therapy resistance and (2) to investigate the hypothesis that "hallmark co-targeting" of mechanisms underlying quasi-independent parameters could result in more enduring therapeutic responses and impaired adaptive resistance to hallmark-targeting therapies. Certainly, multi-targeting tumors with drug combinations looks to be the future for many forms of human cancer, and perhaps this refined design principle could add impact to such strategies, as two case studies suggest. Finally, this synthesis has not sought to address the question: will there prove to be additional functional capabilities, enabling phenotypic characteristics, and hallmark-facilitating cells in the TME that have demonstrable generality across the spectrum of human cancers and hence become conceptually incorporated? Time will tell.

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DECLARATION OF INTERESTS

The author declares no competing interests.

SUPPLEMENTAL INFORMATION

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